

PORETSKOYE, Russian Federation

WMO: 276750

Lat: 55.183N

Lon: 46.333E

Elev: 136

StdP: 99.70

Time Zone: 3.00 (E03)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.4	-23.5	-29.2	0.3	-26.2	-26.5	0.3	-23.4	10.8	-7.0	9.5	-6.9	2.1	200	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.4	19.9	29.2	19.0	27.4	18.5	21.4	29.2	20.5	27.3	19.6	25.7	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.8	24.5	18.0	13.1	23.2	17.1	12.4	22.6	62.7	29.2	59.3	27.1	56.3	25.8	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.3	DB	-29.7	33.3	3.4	2.2	-32.1	34.9	-34.1	36.1	-36.0	37.4	-38.4	38.9	(4)
(5)			WB	-29.9	23.3	3.3	1.6	-32.2	24.4	-34.2	25.3	-36.0	26.2	-38.4	27.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-11.4	-9.3	-4.1	5.9	14.9	17.7	20.0	18.5	12.4	4.8	-1.2	-6.5	(6)
(7)		DBStd	12.12	7.08	6.72	4.97	4.93	4.54	4.20	3.60	4.16	4.09	4.68	4.82	6.32	(7)
(8)		HDD10.0	2836	663	539	436	144	10	2	0	1	22	171	336	512	(8)
(9)		HDD18.3	4974	921	773	694	374	124	61	22	48	181	419	586	771	(9)
(10)		CDD10.0	1095	0	0	0	21	162	232	311	265	94	10	0	0	(10)
(11)		CDD18.3	192	0	0	0	0	18	42	74	54	3	0	0	0	(11)
(12)		CDH23.3	1861	0	0	0	2	194	407	707	509	42	0	0	0	(12)
(13)		CDH26.7	593	0	0	0	0	37	125	247	179	4	0	0	0	(13)
(14)	Wind	WSAvg	3.6	4.0	4.1	4.0	3.9	3.5	3.2	2.8	2.8	3.0	3.7	3.9	4.2	(14)
(15)	Precipitation	PrecAvg	515	26	20	21	36	39	65	72	61	52	48	39	29	(15)
(16)		PrecMax	708	65	62	68	102	118	135	149	208	131	150	70	54	(16)
(17)		PrecMin	356	4	0	2	6	1	23	1	0	6	0	7	2	(17)
(18)		PrecStd	97	14	13	12	24	28	29	38	37	31	29	15	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	2.1	8.2	23.2	29.8	32.1	35.8	27.2	18.4	11.5	4.2	(19)	
(20)			MCWB	0.5	0.7	5.7	13.4	16.8	19.9	21.5	22.8	15.9	13.0	9.1	3.4	(20)
(21)		2%	DB	0.7	1.2	4.9	20.4	27.6	29.9	31.7	31.0	24.2	15.4	9.4	2.5	(21)
(22)			MCWB	0.0	0.1	2.4	11.7	16.3	19.4	20.6	19.7	15.3	11.3	7.8	1.7	(22)
(23)		5%	DB	-0.1	0.5	3.2	17.3	25.7	27.8	29.3	28.1	21.8	13.2	7.1	1.5	(23)
(24)			MCWB	-0.7	-0.4	1.2	10.2	15.8	19.0	19.8	19.0	14.4	10.2	6.0	0.7	(24)
(25)		10%	DB	-1.4	-0.6	2.1	14.4	23.3	25.5	27.2	26.0	19.3	11.3	5.1	0.7	(25)
(26)			MCWB	-2.1	-1.5	0.6	8.7	14.7	17.5	19.1	18.3	13.6	9.0	4.2	-0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.1	6.0	14.6	19.1	21.7	23.1	22.9	17.6	13.6	9.7	3.4	(27)
(28)			MCDWB	1.2	1.8	7.1	20.6	25.6	28.7	31.6	35.3	23.3	17.0	11.2	3.9	(28)
(29)		2%	WB	0.2	0.3	2.6	12.8	17.6	20.6	21.7	21.1	16.3	11.9	7.9	1.9	(29)
(30)			MCDWB	0.7	1.1	4.1	18.4	25.0	27.8	29.9	28.4	21.9	14.7	9.3	2.4	(30)
(31)		5%	WB	-0.7	-0.3	1.5	10.9	16.7	19.8	20.7	19.9	15.3	10.5	6.0	0.8	(31)
(32)			MCDWB	-0.1	0.5	2.7	16.8	24.0	26.2	27.6	26.3	19.9	12.6	7.0	1.3	(32)
(33)		10%	WB	-2.0	-1.5	0.7	9.1	15.5	18.5	19.7	18.9	14.2	9.2	4.3	0.0	(33)
(34)			MCDWB	-1.3	-0.6	2.0	14.0	22.2	24.0	26.0	24.0	18.3	11.1	5.0	0.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	6.3	6.9	8.9	10.9	10.2	10.5	10.4	8.9	5.9	4.2	4.6	(35)
(36)			MCDBR	4.0	4.8	6.8	13.6	14.0	13.2	13.8	13.7	13.7	8.7	5.2	3.6	(36)
(37)			MCWBR	3.8	4.3	5.1	6.9	6.1	5.5	5.5	5.3	6.2	5.6	4.6	3.6	(37)
(38)		5% WB	MCDWR	4.0	4.5	6.2	12.5	12.5	11.6	12.0	11.9	11.3	7.5	4.8	3.7	(38)
(39)			MCWBR	3.8	4.2	4.9	6.6	6.2	5.4	5.7	5.3	5.7	5.5	4.4	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.288	0.322	0.358	0.414	0.388	0.382	0.403	0.427	0.376	0.352	0.318	0.289	(40)	
(41)		taud	2.083	2.047	2.025	2.117	2.319	2.368	2.315	2.232	2.359	2.330	2.223	2.072	(41)	
(42)		Ebn at Noon	607	722	789	801	854	864	835	783	780	701	577	504	(42)	
(43)		Edh at Noon	65	99	128	136	118	114	118	121	93	76	58	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.72	1.66	3.21	4.14	5.49	5.79	5.77	4.55	2.83	1.34	0.54	0.40	(44)	
(45)		RadStd	0.10	0.20	0.31	0.42	0.38	0.52	0.53	0.43	0.39	0.22	0.10	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-167	N/A	N/A

Nomenclature: See separate page